AUTONOMOUS MOBILE ROBOTS · PERCEPTION PIPELINE

MULTI-CAMERA + 3D LIDAR

Nova Carter in Isaac Sim with ROS 2 and YOLO

End-to-end perception pipeline integrating twin stereo cameras with a 3D LiDAR for spatial object detection, ROI point-cloud association, metric XYZ localization, and multi-view consensus.

ROS 2 STACK	SIMULATION & MODEL
Jazzy · TF2 · vision_msgs	**Isaac Sim · YOLOv8n CPU**

▲
Validated on closed trajectory · Base frame: base_link



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ONE PIPELINE FROM TWO CAMERAS AND LIDAR TO FUSED 3D DETECTIONS

PARALLEL CAMERA BRANCHES

Left RGB → YOLO → Detection2DArray → Camera-LiDAR Fusion

vision_msgs/Detection3DArray

Right RGB → YOLO → Detection2DArray → Camera-LiDAR Fusion

vision_msgs/Detection3DArray

LIDAR PROJECTION

3D LiDAR → PointCloud2 → TF → projection into both camera images

FUSED OUTPUT

Left + right → Multi-Camera Fusion

/fusion/final/detections_3d

Trajectory: Trajectory Controller → `/cmd_vel` → Nova Carter · **Outputs:** CSV trajectory · CSV detections · annotated video

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IMAGE SENSING LIMITATION

CAMERA-ONLY YOLO DETECTS OBJECTS BUT CANNOT MEASURE DISTANCE

INPUT STREAM

Front-left RGB camera via ROS 2 · 640 × 360 · YOLOv8n live detection

INFERENCE LATENCY

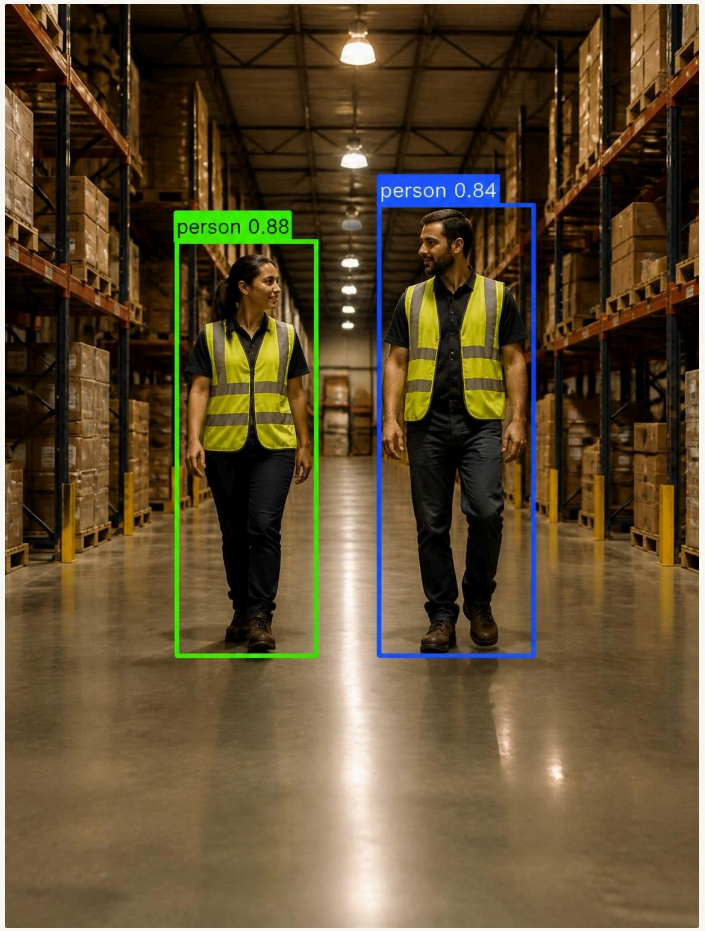
CPU inference about 23–30 ms per frame in tests

ROS 2 OUTPUTS

Publishes `/yolo/detections` (Detection2DArray: class, confidence, 2D box) and `/yolo/annotated_image`

LIMITATION

Reliable image-space localization, but no direct metric 3D distance



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SENSOR COMPLEMENT

LIDAR ADDS THE METRIC DEPTH THAT RGB LACKS

TOPIC: /front_3d_lidar/lidar_points · SENSOR_MSGS/POINTCLOUD2

STEP 01

3D LIDAR XYZ

Raw point cloud acquired from sensor_msgs/PointCloud2 in sensor frame

STEP 02

TF TRANSFORM

Rigid body rotation and translation to camera optical coordinate system

STEP 03

PINHOLE PROJECTION

Intrinsic camera matrix transforms 3D rays into 2D pixel coordinates [u, v]

STEP 04

YOLO ASSOCIATION

Spatial filtering isolating points falling within 2D bounding box ROI

STEP 05

METRIC ESTIMATE

Median filtering extracts robust target XYZ position and Euclidean range

INTRINSIC PINHOLE CAMERA PROJECTION

$$u = f_x \cdot \frac{X}{Z} + c_x$$

$$v = f_y \cdot \frac{Y}{Z} + c_y$$

Coordinate Filtering Constraints:

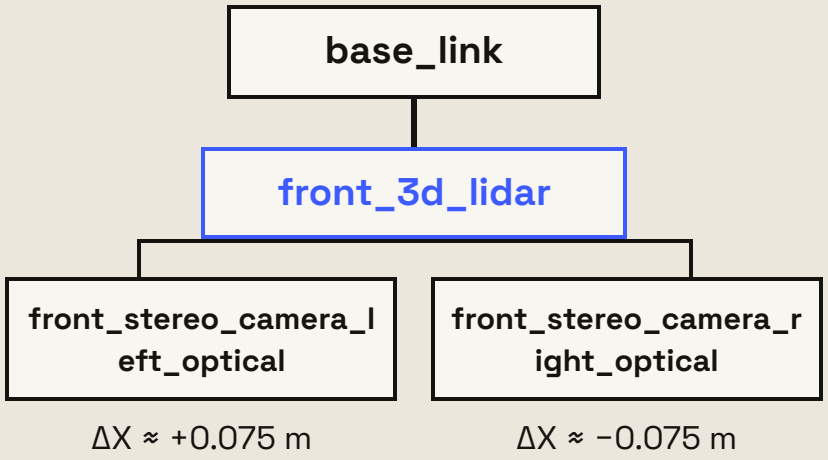
Only points where $Z > 0$ ahead of the image plane and $0 \leq u < W, 0 \leq v < H$ map into the camera image canvas.

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TRANSFORM TREE

TF PUTS LIDAR AND BOTH CAMERAS IN ONE COORDINATE FRAME

OPTICAL CONVENTION: X RIGHT · Y DOWN · Z FORWARD · FINAL 3D DETECTIONS EXPRESSED IN BASE_LINK



EXTRINSIC BASELINE SANITY CHECK

LiDAR sits precisely centered between camera lenses. Measured extrinsic translation: **+75 mm** to left optical center, **−75 mm** to right optical center, confirming mirrored baseline geometry.

OPTICAL VS ROBOTIC COORDINATES

Camera Optical Frame: X right, Y down, Z forward.
Robot Base Frame: X forward, Y left, Z up. TF2 dynamically handles rotations so fused obstacles populate Navigation in base_link.

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ROI ASSOCIATION & DEPTH EXTRACTION

PROJECTED LIDAR POINTS INSIDE EACH YOLO BOX GIVE OBJECT XYZ AND RANGE

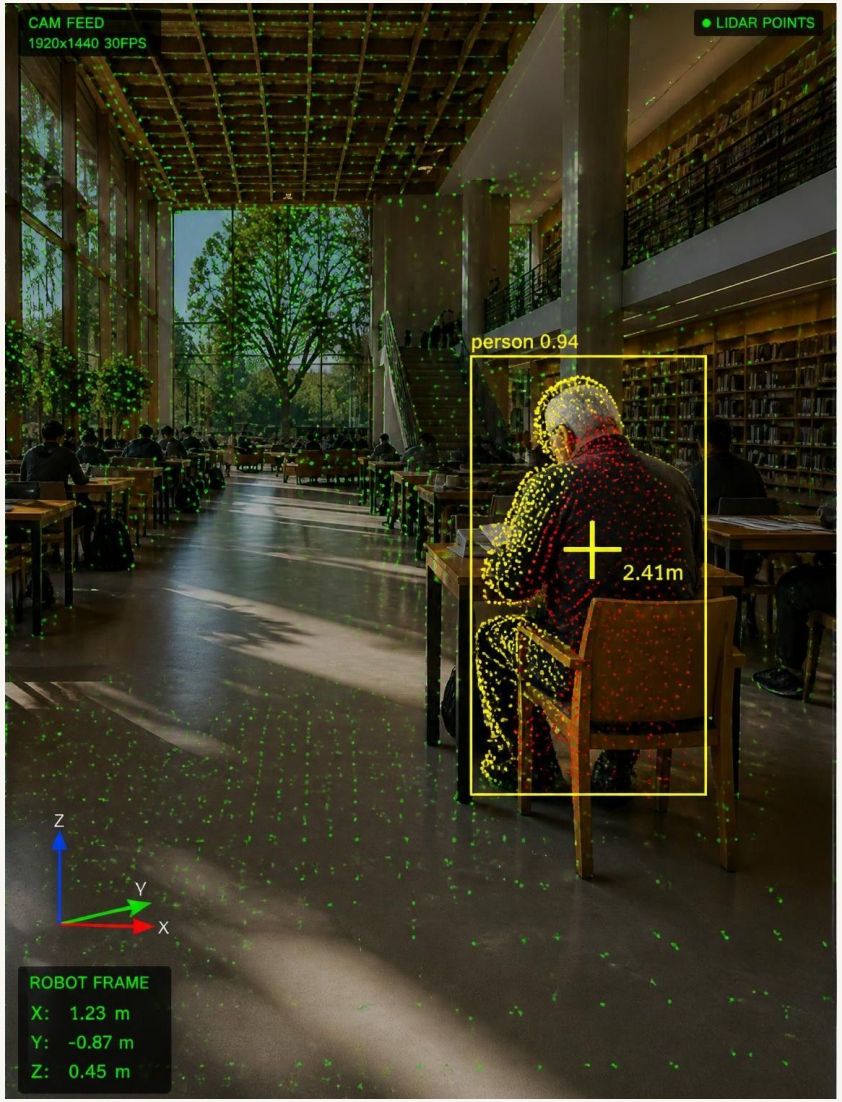
TOPIC: /fusion/detections_3d · VISION_MSGS/DETECTION3DARRAY

Find projected LiDAR points inside each YOLO box; take median XYZ, transform to `base_link`, compute range, and publish.

LIVE DETECTION TELEMETRY (SAMPLE RUN) · Detection3DArray

Person	Confidence · XYZ · Range
Person #1	0.88 · X: 2.41 m, Y: 0.32 m, Z: −0.05 m · 2.43 m
Person #2	0.82 · X: 3.84 m, Y: −0.78 m, Z: −0.02 m · 3.92 m

Future refinement: overlapping ROIs, depth clustering



Green: projected LiDAR · Colored: ROI points · Boxes: YOLO · Yellow cross: median 3D estimate

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REDUNDANCY & TOPIC MAP

A SECOND CAMERA ADDS REDUNDANCY WHEN ONE VIEW MISSES

LEFT STEREO CAMERA	RIGHT STEREO CAMERA
RGB · 640 × 360	**RGB · 640 × 360**
/front_stereo_camera/left/image_raw	/front_stereo_camera/right/image_raw
Camera info	**Camera info**
/front_stereo_camera/left/camera_info	/front_stereo_camera/right/camera_info
Independent YOLO	**Independent YOLO**
/yolo/detections	/yolo/right/detections
Independent fusion	**Independent fusion**
/fusion/detections_3d	/fusion/right/detections_3d

Redundancy: Weak, occluded, or missed in one camera can still be observed by the other; right camera sometimes detected one person instead of two.

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CONSENSUS & ASSOCIATION

MULTI-CAMERA ASSOCIATION MERGES LEFT AND RIGHT IN BASE_LINK

/fusion/final/detections_3d · Detection3DArray

Association rule

Match same class + nearby XYZ (3D Euclidean distance); choose the closest valid pair.

Fuse XYZ by confidence-weighted average; keep unmatched detections from either camera.

Redundancy—not a requirement that every camera detects every object every frame.

CONCEPT

Left camera sees A + B

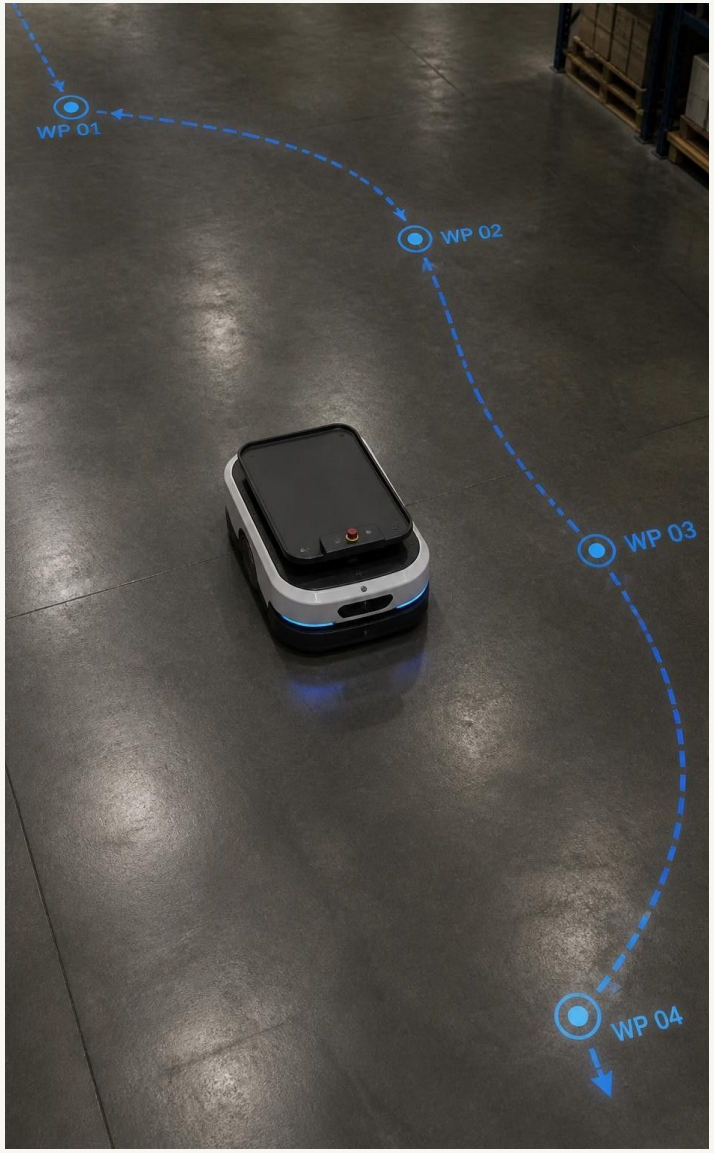
Right camera sees A

Association in base_link

Final output A + B

/fusion/final/detections_3d · Detection3DArray

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Trajectory verification loop in NVIDIA Isaac Sim

DYNAMIC VALIDATION

PERCEPTION KEEPS RUNNING WHILE THE ROBOT FOLLOWS A SET PATH

INPUT: /chassis/odom · OUTPUT: /cmd_vel

Path relative to initial pose

Start	**0.5 m forward**	**0.5 m left**	**1.0 m backward**

During motion: both cameras, YOLO, LiDAR, TF, and multi-camera 3D detections keep running.

> Controlled validation trajectory, not full obstacle-avoidance navigation.

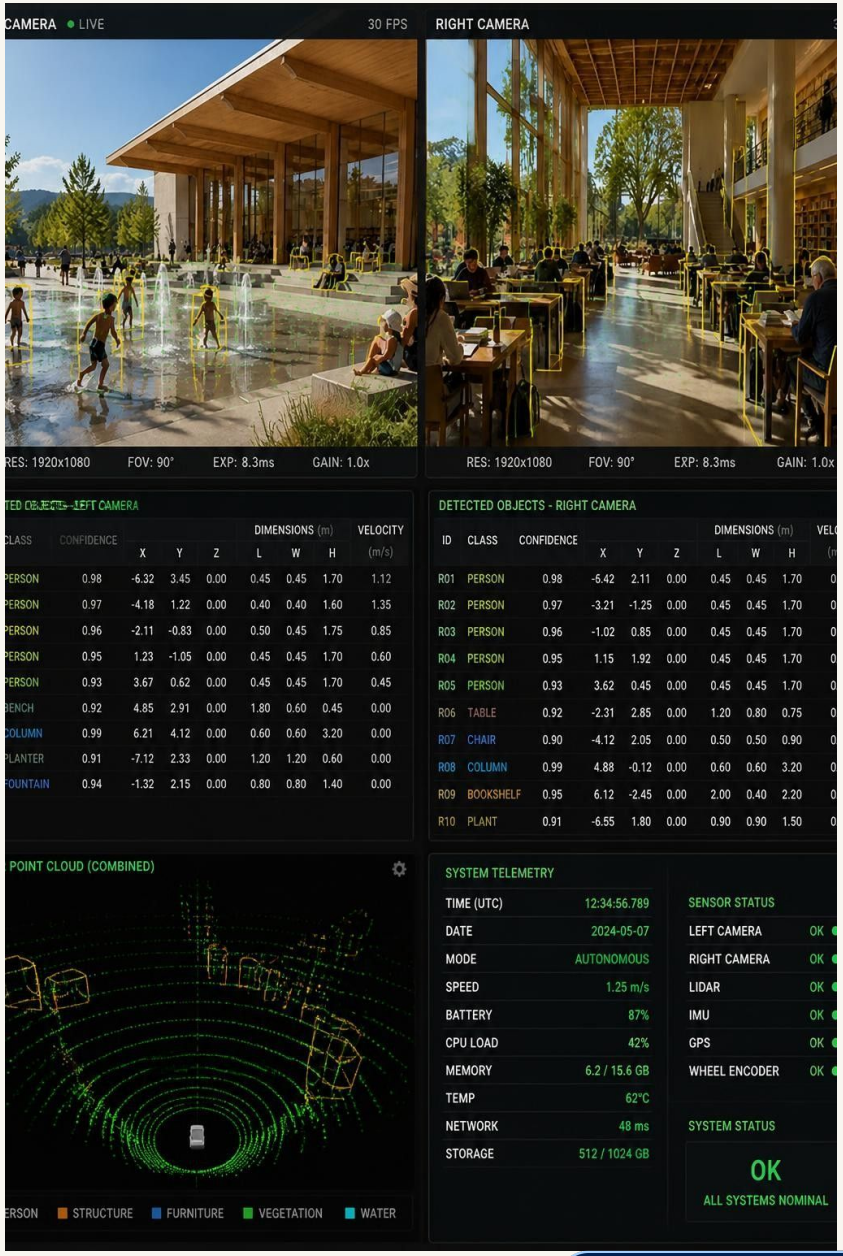
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TELEMETRY & ARTIFACTS

EVERY RUN IS LOGGED AND EXPORTED AS AN ANNOTATED VIDEO

OUTPUT ROOT: outputs/final_demo/run_latest/

LOGGED ARTIFACT	RECORDED FIELDS
trajectory/planned_path.csv **trajectory/executed_path.csv**	Timestamp, X, Y, yaw, linear and angular velocity
detections/detections.csv	Timestamp, object index, class, confidence, X, Y, Z, range
video/perception_motion.mp4	Left and right views side by side with YOLO boxes, projected LiDAR points, 3D estimates



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